

Specification Document of ZigooCare

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1. General Context

1.1. Study of the Existing System and Competing Systems

Currently, waste management and sewer maintenance in most Tunisian municipalities are organized through a combination of manual procedures and digital management tools, based on periodic inspections, administrative reports, and scheduled interventions. Information related to sewer conditions, employees, intervention teams, and recycling stations is generally managed through separate information systems and specialized platforms, each dedicated to a specific operational domain.

1.2. Problem Statement

The problem identified in the current Tunisian waste management framework is a systemic inefficiency driven by fragmentation and a reactive approach to infrastructure maintenance. Currently, municipalities rely on manual inspections and citizen complaints to identify sewer blockages, leading to delayed responses and increased environmental risks. Critically, the accumulation of waste within the sewer network is a primary cause of urban flooding during rainfall; these blockages prevent proper water evacuation, leading to the overflow of polluted water and debris onto public streets. This "reactive" model is further crippled by data silos, where information regarding employees, sewer conditions, and collection logistics is stored in isolated systems. Furthermore, existing international solutions are often ill-suited for the local context; they prioritize fleet logistics over upstream sewer monitoring and fail to provide the predictive analytics necessary for prioritizing urgent interventions in aging infrastructures. Consequently, there is a critical lack of end-to-end traceability, leaving municipalities without the intelligent tools needed to prevent flooding and optimize urban sanitation.

2. Proposed Solution and Its Requirements

As part of improving waste management and recycling in Tunisian urban areas, the ZigooCare project proposes a smart application designed to monitor, organize, and optimize the entire waste management process, from waste detection in sewer systems to final recycling. After authentication, the municipal employee can access detailed information about their municipality, particularly the status of the sewer systems assigned to it. Based on the analysis of data collected from sensors installed in the sewers, the system evaluates their blockage level and overall condition. To ensure proper sewer operation, a collection team composed of external service providers is selected by an authorized agent according to availability and workload. The collected waste is then transported to a compatible recycling center selected based on its processing capacity and availability. This approach ensures intelligent, coordinated, and efficient management of maintenance, collection, and recycling operations.

2.1. Functional Requirements

Table 1: ZigooCare App Modules

Module	Entity	Characteristics	Position	Features
Employee Management	Employee	• employee_id • nic • first_name • last_name • position • email	Personnel Manager	• Basic features: CRUD • Advanced features: 1. Sort the list of users by position

		- password - question - answer - fingerprint		- Search an employee by name - Export list of employees as Excel - Generate statistics of employee distribution by position - Employee Access Rights Management - Password Recovery Using Security Questions
Municipality Management	Municipality	- municipality_id - name - region - address - phone	Municipal Oversight	- Basic features: (CRUD) - Advanced features:

		• email • longitude • latitude • radius		1. Sort Municipality list by region 2. Search Municipality by name 3. Export Municipality data as PDF 4. Generate statistics of municipalities by region 5. Interactive cartographic visualization combining geolocation of municipalities and a heatmap of flooded areas by municipality. 6. Smart Budget Allocation: distributes the ministry's
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				budget fairly based on each municipality's needs, such as sewer problems and maintenance requirements.
Sewer Management	Sewer	• sewer_id • location • max_capacity • current_capacity • water_level • blockage_rate	Ministry Agent	• Basic functionalities (CRUD) • Advanced features: 1. Sort sewer list by blockage rate 2. Search sewer by location 3. Export sewer data as Excel 4. Generate statistics of sewers by blockage rate

				5. AI-based maintenance route optimization using weather conditions, blockage rate, water level, and sewer location 6. The system analyzes historical sewer data to predict future degradation and support preventive maintenance planning.
Collection Teams Management	Collection Team	- team_id - team_size - vehicle_plate - contact_number - status (available / unavailable)	Operations Manager	- Basic functionalities (CRUD) - Advanced features: - Sort list by status

		• price • average_rating • rating_count • active_assignment • average_completion_time		2. Search teams by team id 3. Export team data as PDF 4. Generate team statistics by team status 5. Star rating system: This performance analytics engine automatically ranks teams using a weighted star-rating algorithm to identify top-tier providers and trigger quality audits for underperforming units. 6. The Intelligent Load Balancer; a
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				specialized algorithm that automatically redistributes intervention tasks among active collection teams to ensure that no single team is overwhelmed while others remain underutilized.
Recycling Centers Management	Recycling Center	• center_id • name • address • email • max_capacity • current_capacity • status (active / inactive / saturated)	Recycling Centers Manager	• Basic features: CRUD • Advanced features: 1. Sort the list of recycling centers by status 2. Search recycling

				centers by name
				3. Export recycling centers list as Excel
				4. Generate statistics of recycling centers grouped by capacity intervals
				5. The system automatically sends an email to the center at each assignment of a waste lot. The email includes the new assignment details and the daily totals for that center.
				6. A system prevents the

				selection of a treatment center when its available capacity is insufficient for an intervention. It also displays an alert list of centers close to saturation, showing the number of remaining interventions that can still be handled.
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2.2. Non-Functional Requirements

- **Usability:** The ZigooCare application is designed to be ergonomic and intuitive, with clear and consistent navigation between modules, enabling employees to perform their daily tasks efficiently and with minimal training.
- **Performance:** The application is developed in C++, a low-level programming language known for its efficient resource management and high execution speed, ensuring optimal performance and low system resource consumption.
- **Maintainability:** The system is structured to facilitate maintenance and evolution, allowing future improvements, bug fixes, and feature extensions without impacting existing functionalities.
- **Modularity:** ZigooCare follows an object-oriented and modular architecture, promoting clear separation of concerns, improved code organization, and reusability of components across different modules.
- **Security:** The application implements security mechanisms to protect sensitive data, including encrypted credential storage and protection against SQL injection attacks, ensuring the confidentiality and integrity of user information.

2.3. List of Sustainable Development Goals (SDGs) impacted



Figure 1: SDG 9 Logo

The project supports SDG 9 by using digital tools and data analysis to improve sewer infrastructure management. By predicting failures and degradation, it helps make sanitation systems more reliable and modern.



Figure 2: SDG 11 Logo

This project helps make cities safer and more sustainable by improving sewer system maintenance. Predicting risks in advance reduces flooding, overflows, and health problems, which improves quality of life for citizens.



Figure 3: SDG 13 Logo

The project supports climate action by helping cities adapt to climate change. By anticipating the impact of heavy rain and extreme weather on sewer systems, it reduces environmental damage and pollution.

3. Graphic Design

3.1. Application Logo



Figure 4: ZigooCare Logo

The ZigooCare logo symbolizes smart sewer and waste management in Tunisia. The sewer with flowing water represents monitored drainage systems, while the waste and recycling icons highlight sorting and recycling processes. Green buildings and leaves reflect sustainability, and the Wi-Fi signal emphasizes the smart and connected nature of the system. The map of Tunisia shows the project's local focus. The blue color represents water and reliability, green symbolizes environmental protection and sustainability, and red reflects Tunisian identity.

3.2. Graphic Charter

The graphic charter is based on a balanced palette of blue, green, grey, and neutral tones. Dark blue (#32485F) conveys reliability, technical rigor, and institutional stability, while green (#689C3A) highlights titles and emphasizes sustainability and positive actions. Grey and off-white tones (#8DA2B5 and #DBDBD9) structure the sidebar and navigation areas, ensuring visual comfort and clear separation of interface elements. A translucent background color (#99DBDBD9) is used for content areas to add depth while preserving readability. This palette ensures visual consistency, accessibility, and a clear visual hierarchy across the application. The Onest font is a modern, clean sans-serif typeface designed for digital interfaces, offering excellent readability and a balanced,

contemporary look. Its neutral and structured character makes it well suited for institutional and technical applications, supporting clarity, accessibility, and a professional visual identity.



Figure 5: ZigooCare Color palette

3.3. Graphical User Interfaces (GUI)

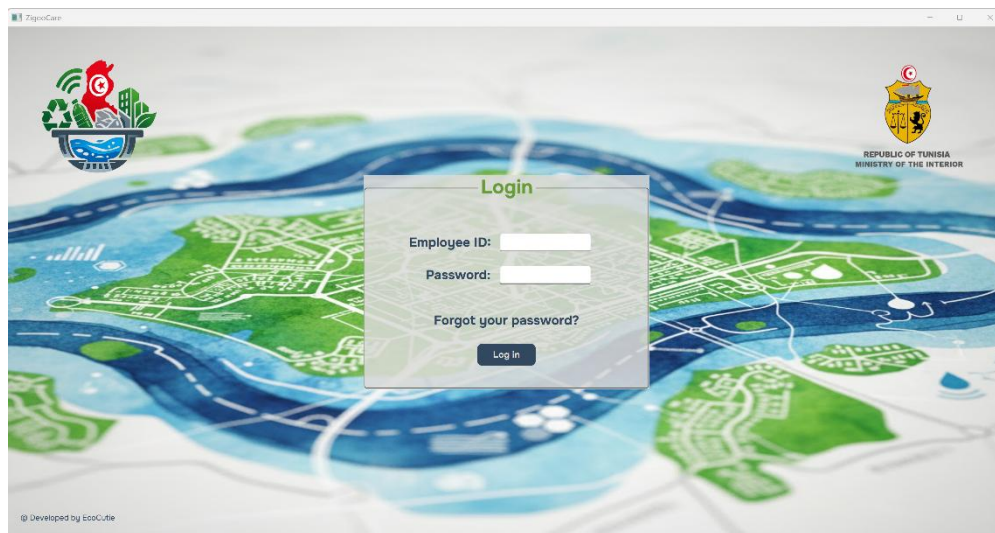


Figure 6: Login interface

Login Interface: Allows employees to access the system by entering their Employee ID and password. Includes options to log in or recover a forgotten password.

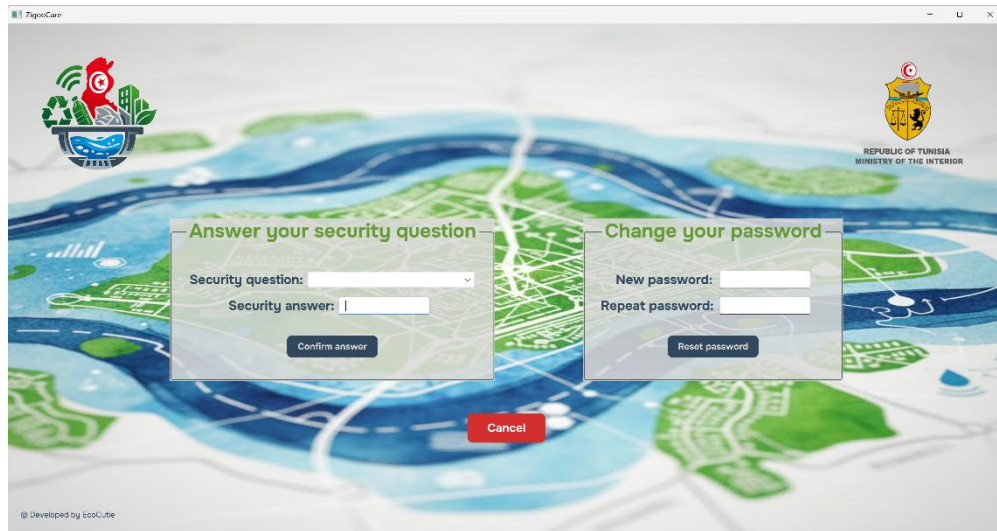


Figure 7: Password reset interface

Password Reset Interface: Allows users to verify their identity by selecting their pre-set security question and providing the correct answer, then securely reset their password by entering a new one twice for confirmation, with options to confirm or cancel.

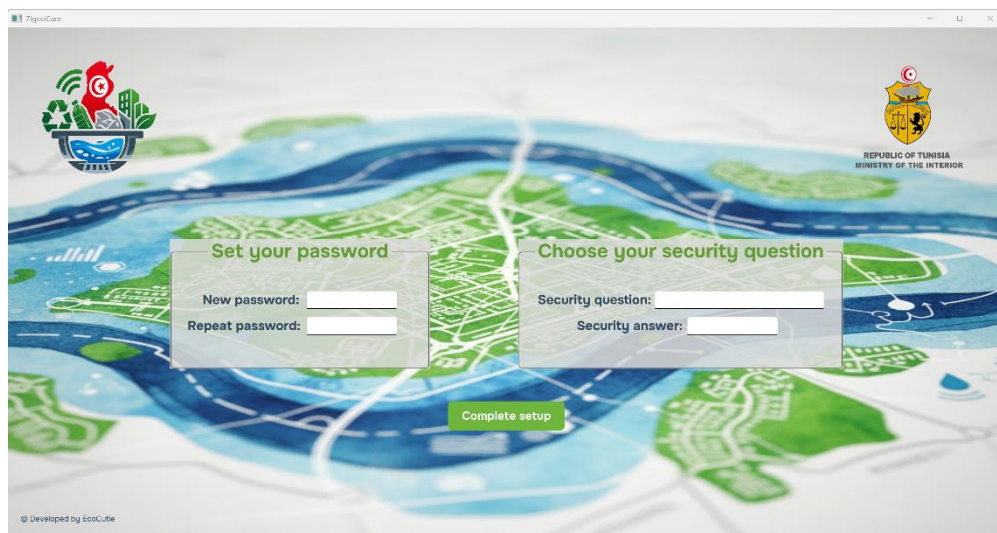


Figure 8: Account setup interface

Account Setup Interface: Guides first-time users to create a password (entered twice) and set a security question with its answer, finalized by the “Complete Setup” button.



Figure 9: Employees dashboard interface

Employee Dashboard Interface: Combines employee statistics and detailed records in a single view, displaying visual workforce indicators alongside a searchable and sortable employee table. It also allows users to add, edit, and manage employee information through an integrated form, with options to confirm changes or export data to Excel.



Figure 10: Municipalities dashboard interface

This dashboard gives you full control over regional data. It features a visual chart showing the number of municipalities per region for quick insights, followed by a complete list where you can search, sort, and export data as a PDF. You can easily add, edit, or delete entries directly. Additionally, use the quick-access buttons to jump to the Interactive Map or the Smart Budget Allocation page.

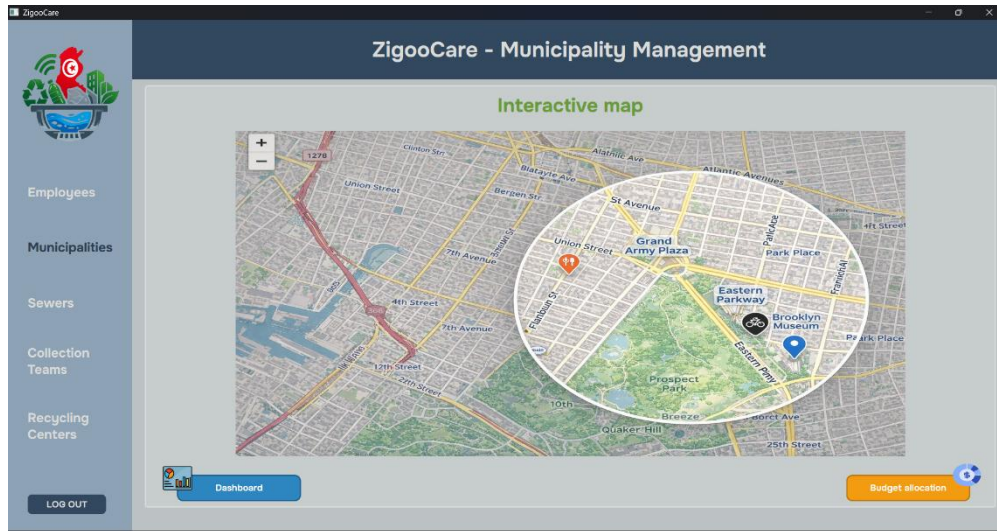


Figure 11: Interactive map interface

This page shows the geographical location of each municipality on an interactive map. High-risk areas are highlighted using different colors based on their global risk level.



Figure 12: Smart budget allocation interface

This page allows users to enter the annual budget. The system then calculates and ranks municipalities in order of priority for budget distribution.



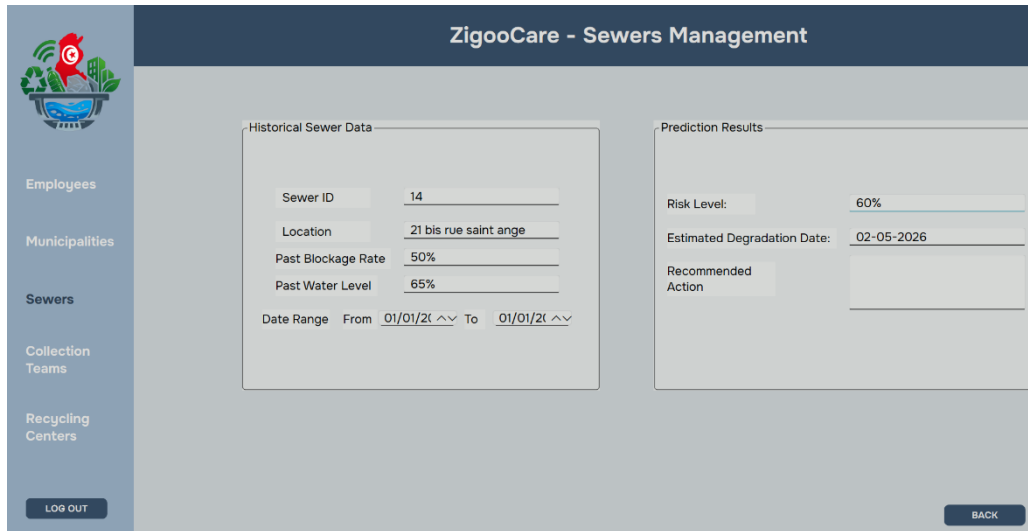
Figure 13: Sewers dashboard interface

This interface, ZigooCare – Sewers Management, provides a centralized dashboard to monitor and manage sewer infrastructure efficiently. It combines a data entry form, a searchable and sortable sewer list, and real-time statistics, including a visual breakdown of sewers by blockage rate. The interface also includes two buttons for advanced functions, SewerAI and PredictIQ, enabling intelligent analysis and predictive decision-making for sewer maintenance.



Figure 14: SewerAI interface

AI-powered interface that optimizes sewer maintenance routes using real-time data and user priorities, showing a map and priority list for faster, more efficient repairs.



The interface is titled "ZigooCare - Sewers Management". It features a sidebar on the left with navigation links: Employees, Municipalities, Sewers, Collection Teams, and Recycling Centers. The main content area is divided into two panels. The left panel, "Historical Sewer Data", contains input fields for Sewer ID (14), Location (21 bis rue saint ange), Past Blockage Rate (50%), Past Water Level (65%), and a Date Range (From 01/01/21 To 01/01/21). The right panel, "Prediction Results", displays Risk Level (60%), Estimated Degradation Date (02-05-2026), and a field for Recommended Action. A "LOG OUT" button is at the bottom left, and a "BACK" button is at the bottom right.

Figure 15: PredictIQ interface

This interface predicts sewer degradation and risk using historical data, providing a risk level, estimated degradation date, and recommended maintenance action to support preventive planning.



The interface is titled "ZigooCare - Collection Team Management". It features a sidebar on the left with navigation links: Employees, Municipalities, Sewers, Collection Teams, and Recycling Centers. The main content area is titled "Collection Team Dashboard". It includes a "Form" section with input fields for Team Size (0), Vehicle Plate, Price, and Status (Available/Unavailable), with a "Confirm" button. Below the form is a "Statistics" section with a pie chart showing "Unavailable" (orange) and "Available" (green) team counts. To the right is a searchable table with columns: Team ID, Team size, Vehicle Plate, Contact Number, Status, and Price. At the bottom right, there are buttons for "Export Team data as PDF", "TeamIQ", and "View Quality Rankings". A "LOG OUT" button is at the bottom left.

Figure 16: Collection teams dashboard interface

This central hub manages the fleet via a CRUD form for team details and a searchable table for data visualization. It features real-time pie chart statistics on team availability and provides navigation buttons to access specialized advanced modules.



Figure 17: Intelligent load balancer interface

This operational engine automates task dispatching by calculating real-time workloads. It features a smart recommendation panel that ranks teams via a "Load Score." By balancing active assignments against completion times, it prevents fleet burnout and ensures rapid response to critical blockages. And a TeamIQ Load Table that integrated data grid provides a granular view of the fleet's current capacity.



Figure 18: Collection teams ranking interface

Acting as a quality audit layer, this screen highlights the "Top 3 Teams" on a leaderboard podium using weighted average ratings. It displays fleet-wide performance via a Star Rating system, helping managers prioritize high-quality teams for upcoming missions.



Figure 19: Recycling centers dashboard interface

A centralized interface that combines recycling center management and data visualization through an integrated form, table, and pie chart.



Figure 20: Saturation prevention interface

A saturation prevention page displaying key metrics on saturated and at-risk recycling centers, with tables highlighting the most critical centers and their capacity and intervention data.



Figure 21: Email settings interface

A page to choose and edit what goes into the automated emails sent to the centers after every intervention.